

# Migration Shocks and Rent Growth in Supply-Constrained Housing Markets

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## Project Motivation and Research Question

The COVID-19 pandemic and the expansion of remote work reshaped where households could live. Instead of treating remote-work adoption itself as the treatment—which is confounded by simultaneous macro shocks—this project studies a clearer causal question: did pandemic-era migration shocks associated with remote-work flexibility contribute to local rent growth, and were these effects stronger in supply-constrained housing markets?

Remote work and regional amenities serve as background mechanisms facilitating migration to attractive areas. The main treatment is pandemic-era population inflow, and the outcome is local rent growth. This framing allows us to estimate the relationship between migration-driven housing demand and local rent growth while exploring heterogeneous effects across regions with differing amenities and housing supply constraints, highlighting which areas experienced the greatest rental pressure.

Main Causal Questions:

1. To what extent are local population inflows associated with rent growth?
2. How is this effect amplified in areas with constrained housing supply?
3. How do background drivers, such as climate and amenities, shape migration patterns and their impact on rent?

## Data

Unit of Analysis: County-year panel for U.S. counties (2018–2019 pre-pandemic; 2020–2023 post-pandemic).

Outcome Variable:

- Local rent growth, measured as annual change in log rental prices.
- Sources: Zillow Observed Rent Index (ZORI); ACS median gross rent (robustness).

Treatment Variable:

- Population inflow or migration-driven housing-demand shock from IRS county-to-county migration data.
- Measures:
  - Annual change in log population:  $\Delta \log(\text{Population}) = \log(\text{Population}_t) - \log(\text{Population}_{t-1})$
  - Net in-migration as share of prior-year population (alternative measure).

#### Control Variables:

- Demographics: population size, age distribution, household composition.
- Economic: median income, income growth, unemployment.
- Housing: renter share, vacancy rate, housing units.
- Sources: ACS, Census Building Permits Survey, WHID.

#### Heterogeneity Variables:

- Housing-supply constraints: low vacancy rate, slow housing growth, limited permits, high renter share.
- Background mechanisms: climate (average temperature) and amenities (parks, cultural sites).

#### Data Accessibility:

All datasets are publicly available: Zillow, IRS migration, ACS, BPS, WHID, Dingel & Neiman work-from-home feasibility dataset.

#### Data Cleaning:

- Missing values imputed using standard techniques.
- Counties with incomplete records excluded.
- All variables aligned to county-year panel for temporal consistency.

## Identification Strategy and Baseline Model

Population growth is endogenous: booming local economies attract residents and increase rents simultaneously. To address this, we exploit pandemic-era migration variation induced by remote-work feasibility and amenity exposure.

#### Baseline Panel Model:

$$\Delta \log(\text{Rent}_{ct}) = \beta \cdot \Delta \log(\text{Population}_{ct}) + X_{ct} \gamma + \alpha_c + \delta_t + \varepsilon_{ct}$$

- $\alpha_c$ : county or metro fixed effects
- $\delta_t$ : year fixed effects
- $X_{ct}$ : time-varying local controls

Exposure Specification: Predict population inflows using pre-pandemic climate/amenity measures interacted with post-2020 period; climate/amenities are exposure variables, not treatments.

## Causal Machine Learning Plan

- Doubly robust / double machine learning methods to flexibly model non-linearities.
- Nuisance models (random forests, gradient boosting) estimate covariate-population-rent relationships without linearity assumptions.
- Outputs: point estimates, confidence intervals, comparison to baseline fixed-effects model.

## Heterogeneity and Robustness

- Test whether population inflows have larger effects in supply-constrained markets using interaction terms.
- Housing supply constraints are treated as heterogeneity factors rather than primary treatments.
- Robustness checks: ZORI vs ACS rent, county-level vs metro-level panels, pre-pandemic trends, exclusion of “superstar” cities, alternative treatment measures (net migration, total population growth, remote-work-predicted migration).

## Expected Contribution

- Provides a clearer causal inference on migration-driven housing demand.
- Quantifies rent elasticity with respect to population inflows.
- Highlights heterogeneous impacts in supply-constrained markets, informing housing policy and affordability research.

## Related Literature

- Mondragon & Wieland (2022, NBER): Remote work and housing demand, motivates migration-focused causal approach.
- Howard (2023, JFE): Short- and long-run effects of remote work on U.S. housing.
- Dingel & Neiman (2020, JPubEcon): Remote-work feasibility, explains cross-county differences in pandemic-era migration.